

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
UNIVERSITY OF INFORMATION TECHNOLOGY



Major: Computer Science
Course: Computational Thinking

Classification of Regional Discriminatory Comments on
Vietnamese Social Media

Students

Full Name	Student ID
Võ Lê Ngọc Thịnh	24521710
Hà Xuân Thiện	24520031
Nguyễn Hồng Phúc	24521390
Vũ Minh Phương	24521421
Trần Quang Trường	24521901

Instructor

PhD. Ngô Đức Thành

Work Assignment

Member	Primary Responsibility	Completion
Trần Quang Trường	Problem Definition	100%
Võ Lê Ngọc Thịnh	Applying CAT to Problem Solving	100%
Nguyễn Hồng Phúc	Breakdown Tree	100%
Vũ Minh Phương	Solution	100%
Hà Xuân Thiện	Evaluation & Ethics & Social Impacts	100%

Note: The responsibilities listed above indicate the primary tasks assigned to each team member. In practice, all members worked collaboratively and also contributed to other aspects of the project whenever necessary.

Contents

1	Problem Definition and Clarification	3
1.1	Problem Statement	3
1.2	Input & Output	3
1.3	Assumptions	3
1.4	Scope	3
1.5	Requirements	4
1.6	Constraints	4
2	Applying CAT to Problem Solving	5
3	Breakdown Tree	6
3.1	Classification of Vietnamese Regional Discriminatory Comments	7
3.2	Text Preprocessing	7
3.2.1	Text Cleaning	8
3.2.2	Text Normalization	8
3.2.3	Word Segmentation	9
3.3	Feature Extraction	10
3.3.1	Text Vectorization	10
3.3.2	Feature Refinement	10
3.4	Text Classification	11
4	Solution	11
4.1	Data Representation	11
4.2	System Overview	12
4.3	Text Preprocessing	13
4.3.1	Text Cleaning	13
4.3.2	Case Normalization	13
4.3.3	De-teencode	14
4.3.4	Word Segmentation	14
4.4	Feature Extraction	15
4.4.1	Text Vectorization	15
4.4.2	Feature Refinement	15
4.5	Text Classification	16

5	Evaluation	16
5.1	Evaluation Dataset	16
5.2	Evaluation Metrics	17
5.2.1	Accuracy	17
5.2.2	F1-Score	17
5.2.3	Latency	17
5.2.4	Mapping Between Requirements and Evaluation Metrics	18
6	Ethical and Social Issues	18

1 Problem Definition and Clarification

1.1 Problem Statement

Existing Vietnamese hate speech research has primarily focused on general forms of abusive language. In contrast, regional discrimination has received far less attention. To bridge this research gap, we propose a solution to detect such comments on social media, aiming to mitigate harmful content and foster a more respectful digital environment.

Within the scope of this problem, regional discrimination is defined as prejudiced comments aimed at judging, degrading, excluding, stereotyping, or exhibiting unjustified favoritism towards an individual or a group of people based on their origin or place of residence.

1.2 Input & Output

To achieve the aforementioned objectives, the system operates based on the following standard specifications:

- **Input:** A raw text string representing a comment extracted from Vietnamese social media platforms.
- **Output:** The predicted label for the input text, categorized into two label types:
 - **Regional Discriminatory:** Prejudiced comments aimed at judging, degrading, excluding, stereotyping, or exhibiting unjustified favoritism towards targets based on their regional origin or place of residence.
 - **Other:** All other comments that do not meet the criteria for label Regional Discriminatory.

1.3 Assumptions

These are the prerequisites that the project team assumes to be true for the system to operate as designed:

- The annotation process follows the labeling guideline. Each comment is labeled according to the agreed definition of regional discrimination, and the final label is used as the ground truth for training and evaluation.
- Each individual comment must be self-contained, meaning that regional discrimination can be identified from the comment text alone without relying on external context.

1.4 Scope

This section defines the boundaries of the project, including what the system will address and what lies beyond its processing capabilities:

In-scope

- **Task:** Binary classification (Regional Discriminatory vs. Other).
- **Domain:** Regional discriminatory content in Vietnam.
- **Input Data:** Individual text-based comments (Facebook, TikTok, Reddit)

Out-scope

- **Author Identity:** Tracking or identifying comment authors.
- **Explainability:** Analyzing the reasoning behind classifications.
- **Other Biases:** Handling gender, race, or religious discrimination.
- **Updateability:** New discriminatory terminology may be missed over time.

1.5 Requirements

Functional Requirements

- The solution shall be able to classify an input text sequence into one of two labels based on the core definition of regional discrimination.

Non-Functional Requirements

To ensure the best user experience as defined by the problem statement, the system must achieve the following threshold indicators:

- **Model Performance:** Achieve an F1-score of at least 90% and an Accuracy of at least 90%.
- **Processing Latency:** The total time for preprocessing and prediction shall not exceed 1 second per comment.

1.6 Constraints

Technical Constraints

- The input data must be plain text encoded in UTF-8.
- Multimedia components such as images, videos, or audio are not allowed.
- The length of each comment must be within the range of 3 to 512 characters.
- At least 70% of the words in the input comment must be written in Vietnamese.
- The input comment may contain linguistic characteristics commonly found in Vietnamese social media comments, including slang, abbreviations, and teencode.

Legal Constraints

- The input comment must not contain personally identifiable information, such as usernames, account IDs, or similar identifying information.

2 Applying CAT to Problem Solving

This section describes how Computational Thinking (CAT) is systematically applied to transform the raw social media comment classification problem into a structured NLP classification pipeline. The process involves six sequential steps, alternating between abstraction, decomposition, and pattern recognition.

Step 1: Abstraction of Problem P

- **Operation:** Raw social media comments are highly unstructured, containing noisy elements, inconsistent casing, and informal teencode. Instead of processing this chaotic linguistic data directly, the input is abstracted into a standardized sequence of meaningful tokens, and eventually into numerical vectors. At the output level, the presence of regional discrimination is represented as a discrete binary label.
- **Result:** The real-world problem P is transformed into an abstract problem P' : a binary text classification problem on numerical feature vectors to predict the target label $Y \in \{0, 1\}$.

Step 2: Decomposition of Problem P'

- **Operation:** To transform raw text into a final prediction, the end-to-end binary classification task cannot be solved in a single operation. It must be broken down into sequential processing stages aligned with the data transformation flow.
- **Result:** Problem P' is decomposed into three sequential subproblems: P_1 (*Text Preprocessing*) to standardize the input text, P_2 (*Feature Extraction*) to convert text into numerical representations, and P_3 (*Text Classification*) to output the final label.

Step 3: Decomposition of Subproblem P_1 - Text Preprocessing

- **Operation:** To systematically prepare the raw data for modeling, the high-level task of standardizing text cannot be executed as a single computational block. It must be divided into granular functions that sequentially address noise removal, format consistency, and linguistic boundaries.
- **Result:** Subproblem P_1 , denoting Text Preprocessing, is decomposed into $P_{1.1}$ (*Text Cleaning*) for removing irrelevant noisy elements, $P_{1.2}$ (*Text Normalization*) for ensuring a consistent textual representation, and $P_{1.3}$ (*Word Segmentation*) for combining multi-syllable syllables into single lexical units.

Step 4: Decomposition of Subproblem P_2 - Feature Extraction

- **Operation:** Following text standardization, the process of extracting numerical features requires transforming tokens into computational spaces and subsequently filtering the most relevant signals for the classifier.
- **Result:** Subproblem P_2 , denoting Feature Extraction, is decomposed into $P_{2.1}$ (*Text Vectorization*) for generating contextual semantic representations, and $P_{2.2}$ (*Feature Refinement*) for deriving the final dense feature vector used by the classifier.

Step 5: Decomposition of Subproblem $P_{1.2}$ - Text Normalization

- **Operation:** Within the preprocessing stage, standardizing the text involves addressing two entirely distinct linguistic anomalies: inconsistent capitalization and the prevalence of internet slang or teencode as specified in the system constraints.
- **Result:** The Text Normalization subproblem is further decomposed into two distinct leaf nodes: $P_{1.2.1}$ (*Case Normalization*) for handling capitalization, and $P_{1.2.2}$ (*De-teencode*) for mapping slang to standard dictionary words.

Step 6: Pattern Recognition - Matching Problem P_3

- **Operation:** It is recognized that comments exhibiting regional discrimination share similar lexical and semantic features, such as discriminatory keywords or specific syntactic structures. Consequently, their refined feature vectors will cluster closely together in the high-dimensional latent space.
- **Result:** The Text Classification subproblem, denoted as P_3 , is matched with supervised machine learning algorithms that excel at finding decision boundaries in high-dimensional spaces. The refined feature vectors are mapped to the final predicted label using an optimal classification algorithm.

3 Breakdown Tree

Figure 1 illustrates the hierarchical decomposition of the problem. Starting from the overall problem of classifying regional discriminatory comments, the task is progressively decomposed into smaller and more manageable subproblems. The hierarchy consists of three main stages: text preprocessing, feature extraction, and text classification. Each stage is further divided into specialized subtasks that collectively transform the raw input comment into the final predicted label.

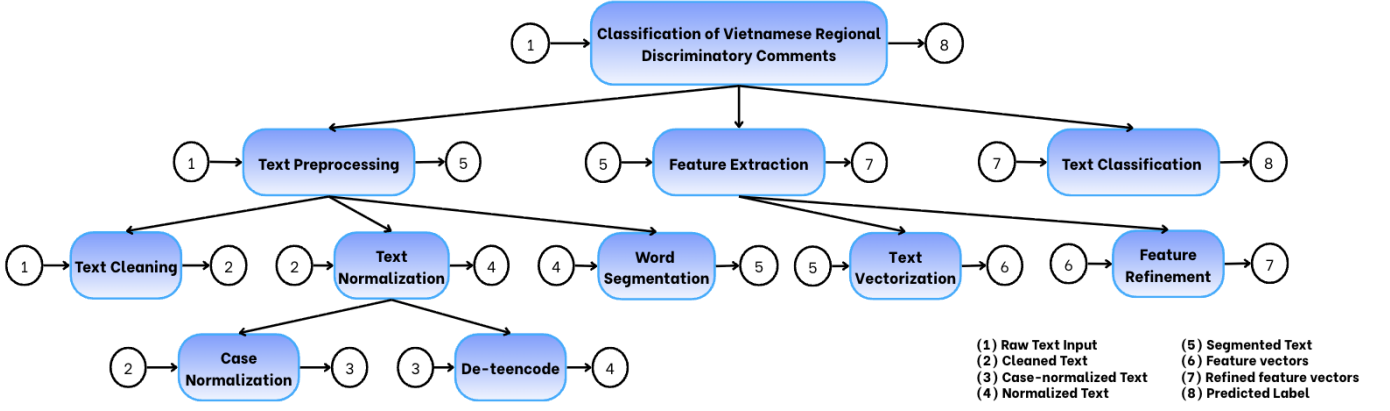


Figure 1: Breakdown tree of the problem.

3.1 Classification of Vietnamese Regional Discriminatory Comments

Classification of Vietnamese Regional Discriminatory Comments is the root node of the decomposition tree and represents the overall problem of identifying regional discriminatory content in Vietnamese social media comments.

Position in the Tree

This node is the highest level in the decomposition tree. It is decomposed into three major subproblems: *Text Preprocessing*, *Feature Extraction*, and *Text Classification*. Together, these subproblems progressively transform the raw input comment into its final predicted label.

Problem Description

The objective of this problem is to determine whether an input Vietnamese social media comment contains regional discriminatory content according to the definition presented in Chapter 1. The input consists of a raw text string extracted from a Vietnamese social media platform. Based solely on the textual content of the comment, the system must assign one of two predefined class labels: *Regional Discriminatory* or *Other*. Therefore, the problem is formulated as a binary text classification task.

Input: (1) **Raw Text Input**, the original Vietnamese social media comment provided as the system input.

Output: (8) **Predicted Label**, the final classification result indicating whether the input comment belongs to the *Regional Discriminatory* class or the *Other* class.

3.2 Text Preprocessing

Text Preprocessing is responsible for transforming raw Vietnamese social media comments into a standardized textual representation suitable for subsequent processing.

Position in the Tree

Text Preprocessing is a child node of *Classification of Vietnamese Regional Discriminatory Comments*. It is further decomposed into four subproblems: *Text Cleaning*, *Case Normalization*, *De-teencode*, and *Word Segmentation*. Together, these subproblems progressively standardize the input comment before feature extraction.

Problem Description

Vietnamese social media comments often contain noisy characters, inconsistent capitalization, informal writing styles, abbreviations, and unsegmented words. These characteristics increase the complexity of subsequent processing and may reduce the quality of the extracted textual representations. Therefore, the objective of this subproblem is to convert the raw comment into a consistent and standardized textual representation while preserving its semantic meaning.

Input: (1) Raw Text Input, the original Vietnamese social media comment.

Output: (5) Segmented Text, the standardized Vietnamese text after preprocessing.

3.2.1 Text Cleaning

Position in the Tree *Text Cleaning* is the first child node of *Text Preprocessing*.

Problem Description The purpose of this subproblem is to remove irrelevant textual elements that do not contribute to the classification task. Typical examples include URLs, emojis, excessive punctuation, and unnecessary special characters. Removing these elements reduces textual noise while preserving the meaningful content of the comment.

Input: (1) Raw Text Input.

Output: (2) Cleaned Text.

3.2.2 Text Normalization

Text Normalization standardizes the writing style of Vietnamese social media comments so that equivalent expressions share a consistent textual representation.

Position in the Tree *Text Normalization* is a child node of *Text Preprocessing*. It is further decomposed into two subproblems: *Case Normalization* and *De-teencode*. Together, these subproblems convert the cleaned text into a normalized textual representation.

Problem Description Vietnamese social media comments frequently contain inconsistent capitalization, abbreviations, teencode, and other informal writing styles. These variations increase

vocabulary sparsity even though they often express the same meaning. The objective of this subproblem is to standardize these variations while preserving the semantic content of the original comment.

Input: (2) Cleaned Text.

Output: (4) Normalized Text.

3.2.2.1. Case Normalization

Position in the Tree *Case Normalization* is the first child node of *Text Normalization*.

Problem Description

The objective of this subproblem is to eliminate unnecessary variations caused by inconsistent letter capitalization. All alphabetic characters are represented in a consistent lowercase form before the next normalization step.

Input: (2) Cleaned Text.

Output: (3) Case-normalized Text.

3.2.2.2. De-teencode

Position in the Tree

De-teencode is the second child node of *Text Normalization*. It receives the case-normalized text and produces the normalized text representation.

Problem Description

The objective of this subproblem is to convert teencode, abbreviations, and informal Vietnamese expressions into their standard written forms while preserving their original semantic meaning. The resulting normalized text provides a more consistent representation for subsequent processing.

Input: (3) Case-normalized Text.

Output: (4) Normalized Text.

3.2.3 Word Segmentation

Position in the Tree *Word Segmentation* is the final child node of *Text Preprocessing*. It receives the normalized text and produces the segmented text used for feature extraction.

Problem Description Vietnamese is a multi-syllable language in which spaces separate syllables rather than complete words. The objective of this subproblem is to identify meaningful lexical units and represent them as individual words, thereby producing a standardized text representation for the subsequent feature extraction stage.

Input: (4) Normalized Text.

Output: (5) Segmented Text.

3.3 Feature Extraction

Feature Extraction transforms the preprocessed text into a numerical representation that captures its semantic information for the subsequent classification task.

Position in the Tree

Feature Extraction is a child node of the root problem. It is further decomposed into *Text Vectorization* and *Feature Refinement*, which together generate the final feature representation used by the classifier.

Problem Description

Machine learning models cannot directly process textual data. Therefore, the objective of this subproblem is to convert the segmented Vietnamese text into numerical feature vectors that preserve its contextual semantic information while remaining suitable for the classification model.

Input: (5) Segmented Text.

Output: (7) Refined Feature Vectors.

3.3.1 Text Vectorization

Position in the Tree *Text Vectorization* is the first child node of *Feature Extraction*. It converts the segmented text into contextual numerical representations.

Problem Description The objective of this subproblem is to transform the segmented Vietnamese text into numerical feature vectors while preserving the contextual semantic information of the original comment. The resulting feature vectors provide the basis for the subsequent feature refinement process.

Input: (5) Segmented Text.

Output: (6) Feature Vectors.

3.3.2 Feature Refinement

Position in the Tree *Feature Refinement* is the second child node of *Feature Extraction*. It receives the feature vectors generated by *Text Vectorization* and produces the refined feature vectors used for classification.

Problem Description The objective of this subproblem is to derive a compact feature representation that effectively summarizes the semantic information of the input comment. The refined feature vectors serve as the input to the subsequent text classification stage.

Input: (6) Feature Vectors.

Output: (7) Refined Feature Vectors.

3.4 Text Classification

Text Classification is responsible for assigning the final class label based on the refined feature representation produced by the previous stage.

Position in the Tree

Text Classification is the final child node of the root problem. It receives the refined feature vectors generated by *Feature Extraction* and produces the final prediction.

Problem Description

The objective of this subproblem is to determine whether the semantic representation of an input comment corresponds to regional discriminatory content. Based on the refined feature vectors, the system assigns one of two predefined class labels: *Regional Discriminatory* or *Other*.

Input: (7) Refined Feature Vectors.

Output: (8) Predicted Label.

4 Solution

4.1 Data Representation

The proposed system processes Vietnamese social media comments through a sequence of well-defined data objects. Each processing component receives a data object, transforms it into another representation, and passes the result to the subsequent component. This design provides a clear interface between processing stages while maintaining a consistent data flow throughout the pipeline. The data objects exchanged between components are summarized in Table 1.

Table 1: Data objects exchanged between system components.

Object	Description
Raw text	The original Vietnamese social media comment provided as the system input.
Cleaned text	The comment after removing non-meaningful characters and irrelevant textual elements.
Case-normalized text	The cleaned comment converted into a consistent lowercase representation.
Normalized text	The comment after de-teencode normalization, where informal writing styles are converted into standard Vietnamese expressions.
Segmented text	The normalized comment after Vietnamese word segmentation, where meaningful multi-syllable words are identified as single lexical units.
Feature vectors	The contextual vector representations generated from the segmented text for subsequent classification.
Refined feature vectors	The feature vectors after the feature refinement process, serving as the final input to the classification model.
Predicted label	The classification result indicating whether the input comment belongs to the <i>Regional Discriminatory</i> class or the <i>Other</i> class.

4.2 System Overview

The proposed solution consists of three major stages: text preprocessing, feature extraction, and text classification. Given an input Vietnamese social media comment, the preprocessing stage first standardizes the input by removing irrelevant textual elements, normalizing writing styles, and segmenting the text into meaningful lexical units. The resulting segmented text is then transformed into contextual feature representations through the feature extraction stage. Finally, the refined feature vector is forwarded to the text classification stage, where the input comment is classified as either *Regional Discriminatory* or *Other*.

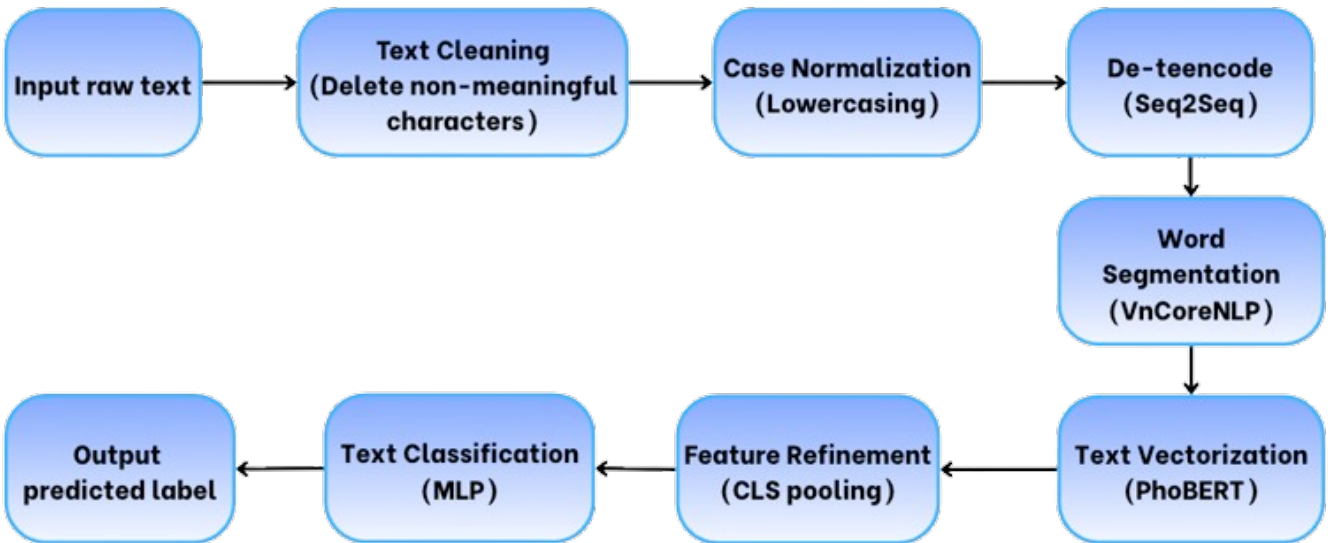


Figure 2: Overall workflow of the proposed solution.

Figure 2 illustrates the overall workflow of the proposed solution. Each stage receives the output of the previous stage and transforms it into a new data representation that serves as the input to the subsequent stage. This modular design simplifies the interaction between system components while ensuring a consistent processing pipeline from the raw input comment to the final prediction.

4.3 Text Preprocessing

Text preprocessing transforms raw Vietnamese social media comments into a standardized textual representation suitable for feature extraction and classification. Since comments collected from social media often contain noisy characters, inconsistent capitalization, teencode, abbreviations, and unsegmented words, directly processing the raw text would reduce the effectiveness of the subsequent feature extraction model.

The preprocessing pipeline consists of four sequential operations: text cleaning, case normalization, de-teencode normalization, and Vietnamese word segmentation. Each operation addresses a specific linguistic characteristic commonly found in Vietnamese social media comments while preserving the semantic information required for classification.

4.3.1 Text Cleaning

The purpose of text cleaning is to remove textual elements that do not contribute meaningful semantic information to the classification task. Social media comments frequently contain URLs, excessive punctuation, repeated special characters, emojis, and other symbols that may introduce unnecessary noise into the feature extraction process.

The cleaning process removes these irrelevant elements while preserving meaningful Vietnamese words and punctuation that may reflect the user’s intention. The output is a cleaned text sequence that contains only the textual information required for subsequent processing.

Algorithm 1 Text Cleaning

Require: A raw Vietnamese social media comment

Ensure: A cleaned text

- 1: remove URLs from the comment
 - 2: remove emojis and non-textual symbols
 - 3: remove redundant punctuation and special characters
 - 4: remove unnecessary whitespace
 - 5: **return** the cleaned text
-

4.3.2 Case Normalization

Vietnamese comments on social media are often written using inconsistent capitalization. The same word may appear in uppercase, lowercase, or mixed-case forms, even though these variations convey the same semantic meaning.

To ensure a consistent textual representation, all alphabetic characters are converted to lowercase.

This normalization reduces unnecessary vocabulary variation and allows semantically identical words to share the same textual representation throughout the processing pipeline.

Algorithm 2 Case Normalization

Require: A cleaned text

Ensure: A case-normalized text

- 1: convert all alphabetic characters to lowercase
 - 2: **return** the normalized text
-

4.3.3 De-teencode

Vietnamese social media users frequently employ teencode, abbreviations, and informal writing styles that differ significantly from standard Vietnamese orthography. These variations increase vocabulary sparsity and may reduce the quality of contextual feature extraction.

To address this issue, the normalized text is transformed into standard Vietnamese expressions through a de-teencode process. Informal words and abbreviated writing forms are converted into their canonical representations while preserving the original semantic meaning. The resulting normalized text provides a more consistent input for the subsequent word segmentation and feature extraction stages.

Algorithm 3 De-teencode

Require: A case-normalized text

Ensure: A normalized text

- 1: identify teencode and abbreviated expressions
 - 2: convert them into standard Vietnamese forms
 - 3: preserve the original semantic meaning
 - 4: **return** the normalized text
-

4.3.4 Word Segmentation

Unlike English, Vietnamese is a multi-syllable language in which spaces separate syllables rather than necessarily separating words. Consequently, meaningful lexical units may consist of multiple adjacent syllables. Processing the text without word segmentation may therefore produce ambiguous or incomplete semantic representations.

The normalized text is segmented into meaningful Vietnamese lexical units using a Vietnamese word segmentation tool. Multi-syllable words are identified and combined into single tokens, producing a segmented text representation that is more suitable for contextual embedding generation by PhoBERT.

Algorithm 4 Word Segmentation

Require: A normalized text

Ensure: A segmented text

- 1: perform Vietnamese word segmentation
 - 2: combine multi-syllable words into single lexical units
 - 3: **return** the segmented text
-

4.4 Feature Extraction

Feature extraction transforms the segmented text into a numerical representation that captures its semantic information for the subsequent classification stage. The proposed solution consists of two steps: text vectorization and feature refinement. Text vectorization generates contextual representations of the input comment, while feature refinement derives the final feature vector used by the classifier.

4.4.1 Text Vectorization

The segmented text produced by the preprocessing pipeline must be transformed into a numerical representation before it can be processed by the classification model. Instead of relying on manually designed linguistic features, the proposed system employs a pretrained language model to automatically learn contextual semantic representations of Vietnamese text.

PhoBERT is adopted as the feature extraction model because it has been pretrained on a large-scale Vietnamese corpus and is capable of capturing both syntactic and semantic relationships between words. Given a segmented Vietnamese comment, PhoBERT encodes the entire text sequence into a set of contextual hidden representations. These representations provide contextual information for every token in the input sequence and serve as the basis for the subsequent feature refinement stage.

4.4.2 Feature Refinement

Among these representations, the hidden state corresponding to the special [CLS] token is used as the representation of the entire input comment. Since this vector summarizes the semantic information of the complete text sequence, it serves as the refined feature vector for the subsequent classification stage.

The extracted feature vectors provide a dense semantic representation of Vietnamese comments while preserving contextual information that cannot be captured by traditional bag-of-words or TF-IDF representations. The resulting refined feature vector is then forwarded to the text classification component.

Algorithm 5 Feature Extraction

Require: A segmented text

Ensure: A refined feature vector

- 1: tokenize the segmented text using the PhoBERT tokenizer
 - 2: encode the token sequence using PhoBERT
 - 3: obtain the contextual hidden representations
 - 4: select the hidden representation of the [CLS] token
 - 5: **return** the refined feature vector
-

4.5 Text Classification

The refined feature vector produced by the feature extraction stage is used to determine whether the input comment contains regional discriminatory content. Since the task involves assigning one of two predefined categories to each input comment, the problem is formulated as a binary text classification task.

The proposed system employs a Multi-Layer Perceptron (MLP) classifier to perform the final prediction. The MLP receives the refined feature vector generated by PhoBERT and learns the mapping between semantic representations and their corresponding class labels using the labeled training dataset.

During system preparation, the classifier is trained using the refined feature vectors extracted from the training comments. During inference, the same feature extraction process is applied to an incoming comment, and the resulting feature vector is forwarded to the trained classifier. The classifier then predicts one of the two predefined labels:

- **Regional Discriminatory:** comments containing regional discrimination according to the problem definition.
- **Other:** comments that do not satisfy the definition of regional discrimination.

The predicted label is subsequently returned as the final output of the proposed system.

Algorithm 6 Text Classification

Require: A refined feature vector

Ensure: A predicted label

- 1: feed the refined feature vector into the trained MLP classifier
 - 2: compute the prediction scores for the predefined classes
 - 3: determine the class with the highest prediction score
 - 4: return the predicted label
-

5 Evaluation

5.1 Evaluation Dataset

The evaluation dataset was collected by crawling the comments from Vietnamese social media platforms such as Facebook, TikTok, and Reddit. After collection, the data were cleaned, anonymized, and manually labeled into two classes: Regional Discriminatory and Others, based on the problem definition.

The dataset includes both normal comments and regional discriminatory comments. In addition, the data also cover slang, abbreviations, loanwords, teencode, and distorted writing forms, in order to evaluate the solution’s ability to handle real-world comments.

5.2 Evaluation Metrics

5.2.1 Accuracy

Accuracy measures the proportion of comments that are correctly classified by the system over the total number of evaluated comments. A prediction is considered correct when the predicted label is the same as the manually annotated ground-truth label.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

5.2.2 F1-Score

Precision measures how many comments predicted as *Regional Discriminatory* are actually discriminatory.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall measures how many actual *Regional Discriminatory* comments are successfully detected by the system.

$$\text{Recall} = \frac{TP}{TP + FN}$$

The F1-score is the harmonic mean of Precision and Recall.

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The terms are defined as follows:

- TP (True Positive): The number of regional discriminatory comments correctly classified as *Regional Discriminatory*.
- FP (False Positive): The number of non-discriminatory comments incorrectly classified as *Regional Discriminatory*.
- FN (False Negative): The number of regional discriminatory comments incorrectly classified as *Other*.
- TN (True Negative): The number of non-discriminatory comments correctly classified as *Other*.

5.2.3 Latency

Latency measures the total time required for the system to process one input comment and return the predicted label. This includes the time for preprocessing, feature extraction, and model inference.

$$\text{Latency} = t_{\text{end}} - t_{\text{start}}$$

where:

- t_{start} : The timestamp when the system receives the input comment.
- t_{end} : The timestamp when the system completes processing and returns the prediction result.

5.2.4 Mapping Between Requirements and Evaluation Metrics

The following table shows the relationship between the system requirements and the evaluation metrics used in this project.

Technical Requirement	Corresponding Metric	Target Value
Model Performance	Accuracy, F1-score	$\geq 90\%$
Processing Latency	Latency	< 1.0 second

Table 2: Mapping between technical requirements and evaluation metrics.

6 Ethical and Social Issues

The proposed solution is intended to assist human moderators in identifying regional discriminatory comments on Vietnamese social media platforms. Although the system aims to promote a more respectful online environment, automatic text classification inevitably involves ethical and social challenges that must be carefully considered during both deployment and practical use.

Several limitations may affect the reliability and fairness of the proposed solution:

- **False Positives:** Neutral, humorous, sarcastic, or context-dependent comments may be incorrectly classified as regional discriminatory, potentially restricting legitimate communication.
- **False Negatives:** Implicit, subtle, or newly emerging discriminatory expressions may not be detected, allowing harmful content to remain on the platform.
- **Bias Risk:** Since the training data are collected from social media, they may contain existing societal biases that could be reflected in the model’s predictions.
- **Human Oversight:** The proposed solution is designed to support human moderators rather than replace them. Classification results should be treated as recommendations and should not be used as the sole basis for deleting comments, suspending accounts, or imposing penalties.
- **Social Impact:** When applied responsibly, the system can help reduce regional discrimination and foster healthier online discussions. However, inappropriate deployment or overreliance on automated predictions may negatively affect users’ freedom of expression.